

Week 2 : Differentiation, Harmonic and Analytic Functions

Def: A function $f: S \subseteq \mathbb{C} \rightarrow \mathbb{C}$ is analytic at $z_0 \in S$ if it is differentiable at every point z in a neighbourhood of z_0 . Furthermore, f is analytic on an open subset $U \subseteq S$ if it is analytic at all points $z_0 \in U$.

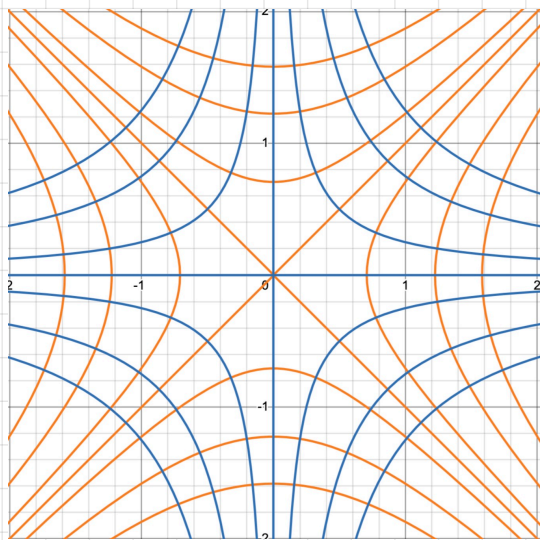
Example 1: $f(z) = z^2 = x^2 - y^2 + 2xyi$

$$u(x, y) = x^2 - y^2 \quad v(x, y) = 2xy$$

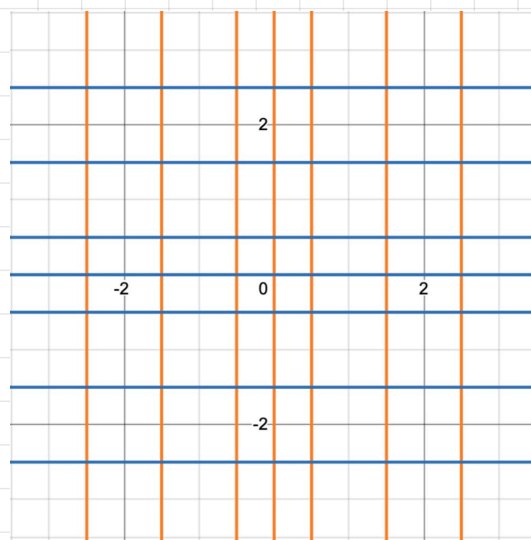
$$\begin{array}{ll} u_x = 2x & v_x = 2y \\ u_y = -2y & v_y = 2x \end{array}$$

Since the partial derivatives are continuous and satisfy the C-R equations, u is analytic for all $z \in \mathbb{C}$.

We also saw:



Inputs.



Outputs

Example 2: $f(z) = \frac{1}{z}, z \neq 0$

$$\frac{1}{x+yi} = \frac{x-yi}{x^2+y^2} = \underbrace{\frac{x}{x^2+y^2}}_u - \underbrace{\frac{yi}{x^2+y^2}}_v$$

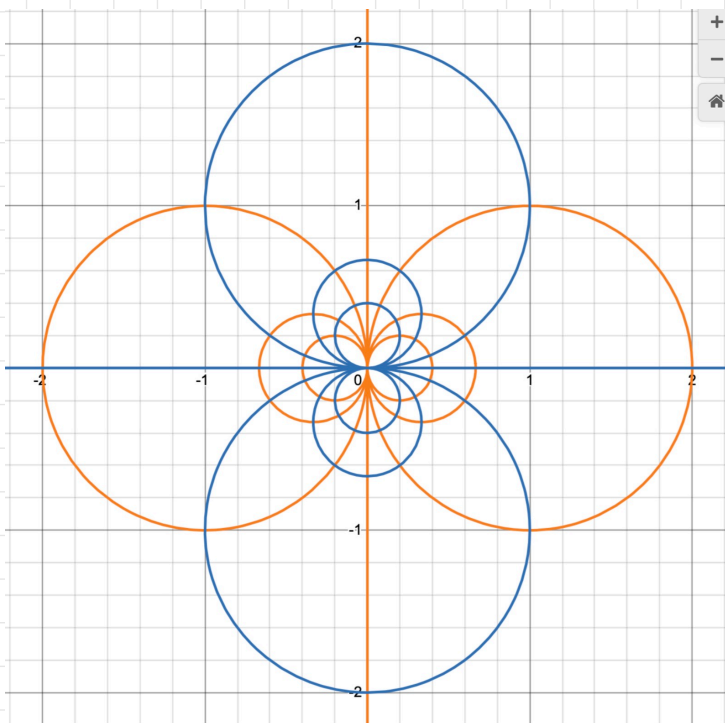
$$u_x = \frac{(x^2+y^2) - 2x \cdot x}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2} \quad v_x = \frac{2xy}{(x^2+y^2)^2}$$

$$u_y = \frac{-2xy}{(x^2+y^2)^2} \quad v_y = \frac{-(x^2+y^2) + 2y \cdot y}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

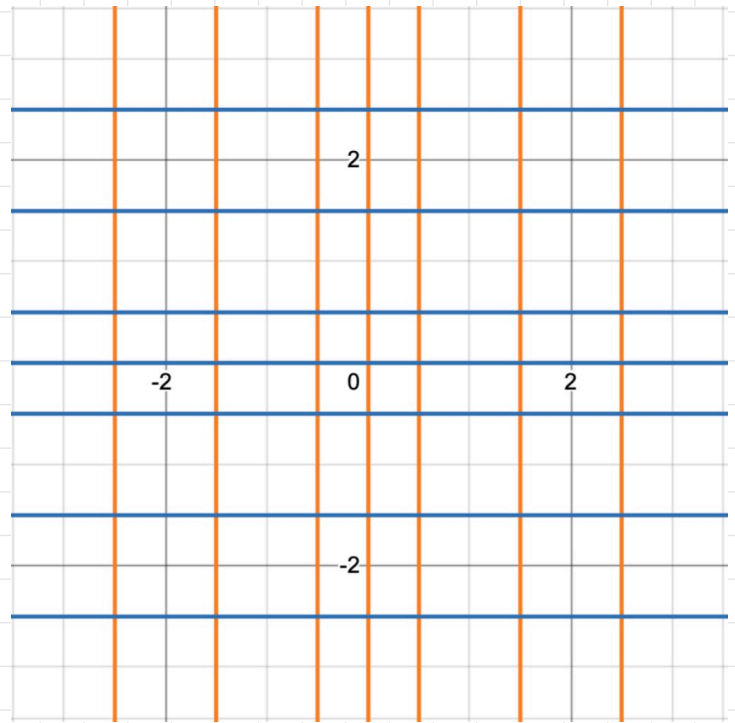
When $(x,y) \neq (0,0)$, the partial derivatives exist and are continuous. Moreover, they satisfy the C-R equations.

Hence, f is differentiable for all $z \neq 0$. That is, it is analytic on $\mathbb{C} \setminus \{0\}$.

Also as a map:



Inputs



Outputs.

Def: A point z_0 is a singular point of f if f fails to be analytic at z_0 , but it is analytic for some point any neighbourhood of z_0 .

A function is called entire if it is analytic on all of $\mathbb{C}$.

Remark: From the differentiation rules we know that a sum and product of analytic functions is analytic.

A quotient $\frac{p(x)}{q(x)}$ of analytic functions is also analytic wherever $q(x) \neq 0$.

Finally, a composition of analytic functions is also analytic.

Examples: • $f(z) = \frac{z^3 + 4}{(z^2 - 3)(z^2 + 1)}$ is analytic on $\mathbb{C} \setminus \{\pm i, \pm \sqrt{3}\}$.

• $f(z) = \cosh x \cos y + i \sinh x \sin y$

$$u(x, y) = \cosh x \cos y$$

$$v(x, y) = \sinh x \sin y$$

$$u_x = \sinh x \cos y$$

$$v_x = \cosh x \sin y$$

$$u_y = -\cosh x \sin y$$

$$v_y = \sinh x \cos y$$

Since the partial derivatives exist, are continuous and satisfy the C-R equations, f is an entire function.

Analytic functions have many amazing properties, which we will spend the rest of the quarter studying. Let's look at some simple ones first.

Thm: If $f'(z) = 0$ everywhere inside a domain D , then $f(z)$ is constant throughout D .

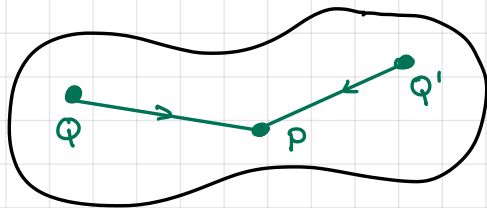
(Note that a domain is a non-empty, open and connected set. And for us, a connected set has all points connected by a polygonal line.)

Proof: Since $f'(z)$ exists for all z we know

$$f'(z) = u_x(x,y) + i v_x(x,y) = 0$$

$$= v_y(x,y) - i u_y(x,y) = 0$$

ie. $u_x = v_x = v_y = u_y = 0$ for all $(x,y) \in D$.



Let $P = (x_p, y_p)$, $Q = (x_q, y_q)$, $Q' = (x_{q'}, y_{q'})$
be points in D s.t.

the straightlines joining P and Q ,
and P and Q' are also in D .

Then the directional derivative of u along PQ is

$$\begin{aligned} D_{PQ} u &= \nabla u \cdot \hat{PQ} \quad \text{where } \hat{PQ} = \text{unit vector} \\ &= 0 \quad \text{joining } P \text{ to } Q \end{aligned}$$

ie. u has a directional derivative of zero at all points on PQ . Hence, u is constant on PQ .

By the same argument u is constant on PQ' . Since the point P is common to both lines, u is equal to the same constant on both lines.

Also since D is a non-empty, open and connected domain, any two points can be joined by a finite number of straight lines. That is, u is constant on D .

We repeat the same argument for v . Hence, $f(z) = c$. $\square$

Corollary: Suppose $f(z)$ and its complex conjugate $\overline{f(z)}$ are both analytic on a domain D .

Then $f(z)$ and $\overline{f(z)}$ are constant on D .

Proof: By the C-R equations for $f(z) = u + iv$:

$$u_x = v_y \quad \text{and} \quad v_x = -u_y \quad (1)$$

By the C-R equations for $\overline{f(z)} = u - iv$:

$$u_x = -v_y \quad \text{and} \quad -v_x = -u_y \quad (2)$$

Combining (1) and (2) we know:

$$u_x = v_y = -v_y \quad \text{and} \quad v_x = -u_y = u_y$$

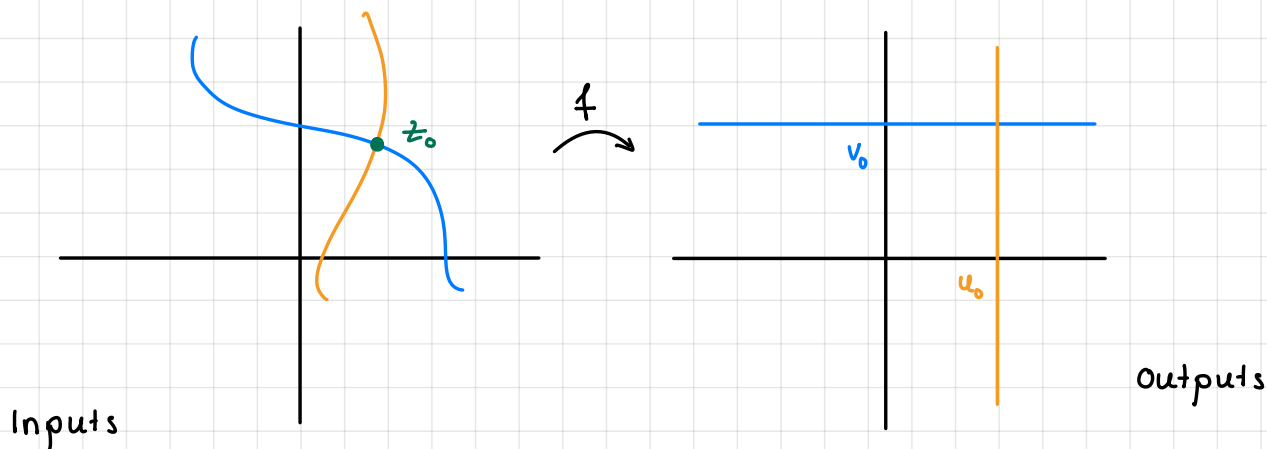
which is only true if $v_y = 0 = u_y$ for all $z \in D$.

But then $f'(z) = 0$ for all z and $f(z)$ is constant. $\square$

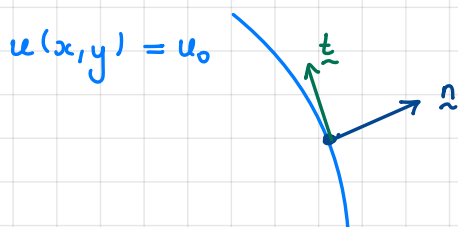
Corollary: If f is analytic in a region D and $|f(z)| = c$ for all $z \in D$, then $f(z)$ is constant on D .

Mappings by Analytic Functions

There are two more interesting properties that come from the Cauchy-Riemann equations. The first has to do with mappings.



For a given point z_0 , $f(z_0) = u_0 + iv_0$. What can we say about the level sets at z_0 ?



The normal vector, in the direction of steepest ascent for the function u , is given by the gradient

$$\nabla u = (u_x, u_y)^T.$$

In particular, the normal vectors to $u(x,y)=u_0$ and $v(x,y)=v_0$ are

$$\underline{n}_u = \nabla u = (u_x, u_y)$$

$$\underline{n}_v = \nabla v = (v_x, v_y)$$

$$\Rightarrow \underline{n}_u \cdot \underline{n}_v = (u_x, u_y) \cdot (v_x, v_y)$$

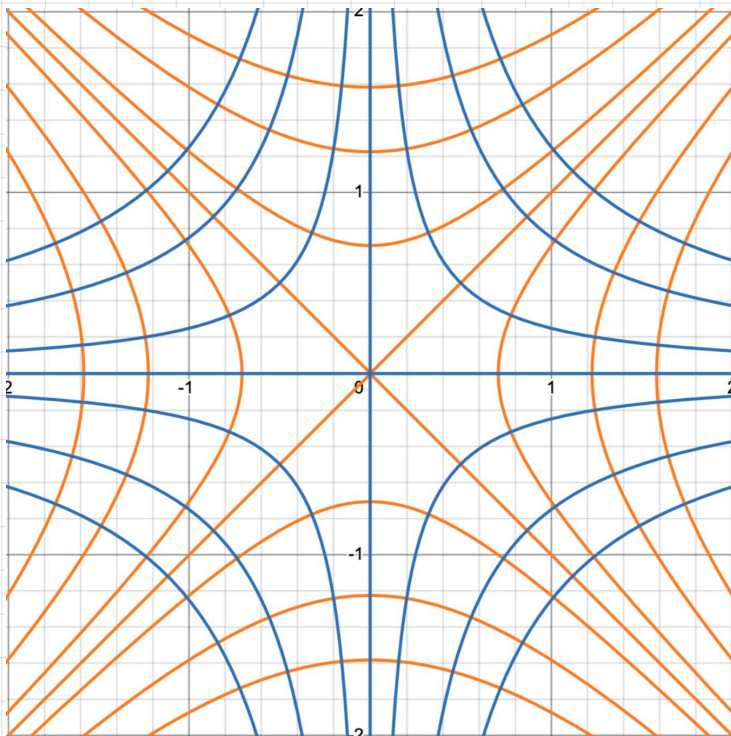
$$= u_x v_x + u_y v_y$$

$$= -u_x u_y + u_y u_x = 0 \quad \text{by C-R equations.}$$

Hence, the normal vectors are perpendicular.

$\Rightarrow$ the tangent vectors are perpendicular.

We saw this above for $f(z) = z^2$.



Question: what is happening at the origin ??

Harmonic Functions

Another consequence of the Cauchy-Riemann equation is that analytic functions are also harmonic functions.

Thm: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic functions in D . That is,

$$u_{xx} + u_{yy} = 0 \quad \text{and} \quad v_{xx} + v_{yy} = 0$$

for all $(x,y) \in D$.

Note that inherent in this statement is that u and v can be differentiated twice, not just once. In fact, analyticity of f guarantees u and v can be differentiated infinitely many times. We will show this in Week 4.

Assuming this fact, we can prove the theorem using only the C-R equation:

$$u_x = v_y \quad \text{and} \quad v_x = -u_y$$

$$\Rightarrow u_{xx} = v_{yx} \quad \Rightarrow v_{xx} = -u_{yx} \quad \text{and}$$

$$\text{and } u_{xy} = v_{yy} \quad v_{xy} = -u_{yy}$$

So using continuity of partial derivatives:

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$$

$$\text{and } v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0$$

Harmonic functions are solutions to Laplace's Equation in 2D:

$$\Delta f = \nabla^2 f = f_{xx} + f_{yy} = 0$$

These functions have special relevance in applied mathematics and physics where they correspond to stationary temperature distributions or electrostatic potentials with 2D geometries.

Mathematically they also have many interesting properties which are deeply related to the properties of analytic functions as conformal maps (see Brown + Churchill Ch 9).

Def: Functions $u(x,y)$ and $v(x,y)$ are harmonic conjugates on D if they satisfy the Cauchy-Riemann equations:

$$u_x = v_y, \quad v_x = -u_y.$$

From this we get a new criteria for analyticity.

Thm: A map $f(z) = u(x,y) + iv(x,y)$ is analytic on D
 $\Leftrightarrow v$ is a harmonic conjugate of u .

From this we might ask: given a function $u(x,y)$, can we find an analytic function f with $\operatorname{Re}(f(z)) = u(x,y)$?

- First u must be harmonic.
- Second we need to find a harmonic conjugate v .

Example: $u(x,y) = y^3 - 3x^2y$.

By differentiating: $u_x = -6xy$ $u_y = 3y^2 - 3x^2$.

Using the Cauchy-Riemann equations:

- $v_y = u_x = -6xy$
 $\Rightarrow v(x,y) = -3xy^2 + \phi(x)$.
- $v_x = -u_y = -3y^2 + 3x^2$
 $\Rightarrow v(x,y) = -3xy^2 + x^3 + \bar{\phi}(y)$.

Combining these relations we see: $v(x,y) = -3xy^2 + x^3 + C$.

The corresponding analytic function is:

$$\begin{aligned} f(z) &= y^3 - 3x^2y + i(-3xy^2 + x^3 + C) \\ &= i(z^3 + C). \end{aligned}$$

Question: Is this always possible?

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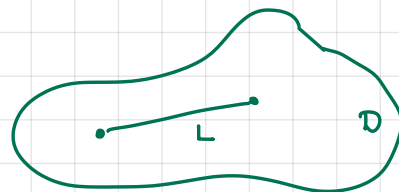
Uniquely Determined Analytic Functions

Analytic functions can be uniquely determined by their values on subsets of their domains. These ideas become particularly relevant when thinking about analytic continuation which we will discuss more next week.

Lemma: Suppose (a) f is analytic on a domain D

(b) $f(z) = 0$ at each point z of a line segment contained in D .

Then: $f(z) \equiv 0$ in D .



We will prove this Lemma later (see C-B Ch6 Section 75) as it requires some additional machinery.

There are some direct consequences of this Lemma.

Corollary:

Suppose $f(z)$ and $g(z)$ are analytic on the domain D and they are equal on a line segment L . Then $f(z) = g(z)$ for all $z \in D$.

Thm: An analytic function on a domain D is uniquely determined by its values on any line segment $L \subseteq D$.

Rmk: In fact, you do not even need a full line segment.
 Suppose you have a sequence $\{z_n\}$ s.t. $z_n \rightarrow z_0$ as $n \rightarrow \infty$
 and $z_n \neq z_0$ for any n . Then

$$f(z_n) = g(z_n) \text{ for all } n \Rightarrow f(z) \equiv g(z) \text{ for } z \in D.$$

These ideas can be used to extend the domain of analytic functions.

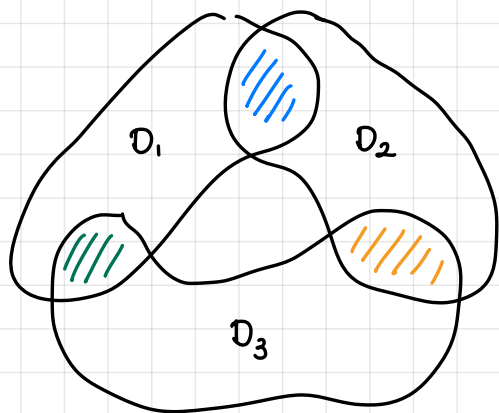
If $f: D_1 \rightarrow \mathbb{C}$ is analytic, then we might be able to find a second function $\tilde{f}: D_2 \rightarrow \mathbb{C}$ which is also analytic and equals f on $D_1 \cap D_2$.

By the above, if such a function exists, it is the unique such choice.

The function
$$F(z) = \begin{cases} f(z) & z \in D_1 \\ \tilde{f}(z) & z \in D_2 \end{cases}$$

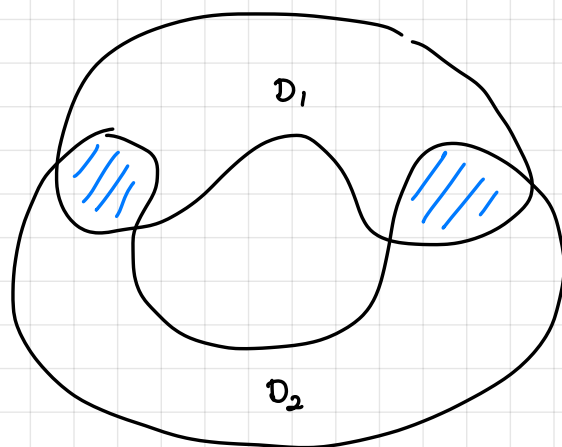
is called the analytic continuation of f onto $D_1 \cup D_2$.

Caution: Care must be taken in some situations.



If $\hat{f}$ is an analytic continuation of f into D_3 , then there is no guarantee that

$$\hat{f}(z) = \tilde{f}(z) \text{ for } z \in D_2 \cap D_3.$$



In this case $D_1 \cap D_2$ is not a domain since it is not connected.

The above theorems do not apply in this case.

Elementary Functions (B+C Ch3)

Now we want to look at some important examples of analytic functions.

Exponential Function

The exponential function for $x \in \mathbb{R}$ is

$$e^x := \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

We can ask whether we can find an analytic function $f(z)$ which agrees with e^x for $z \in \mathbb{R}$. In fact the power series gives us exactly that function

$$e^z := \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

Moreover, using this definition we get Euler's formula

$$f(z) = e^z = e^x e^{iy} = e^x (\cos y + i \sin y)$$

It has the properties:

↖ recall this formula comes from the Taylor series definition of e^x .

$$(1) \quad e^{z_1} e^{z_2} = e^{z_1 + z_2}$$

$$(2) \quad \frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}$$

$$(3) \quad \frac{d}{dz} (e^z) = e^z \quad \text{ie. } e^z \text{ is } \underline{\text{entire}}$$

$$(4) \quad e^z \neq 0 \quad \text{for all } z$$

$$(5) \quad |e^z| = e^x \quad \text{and} \quad \arg(e^z) = y + 2n\pi, \quad n \in \mathbb{Z}.$$

$$(6) \quad e^{z+2\pi i} = e^z \quad \text{ie. } e^z \text{ is periodic with an imaginary period } 2\pi i.$$

Example: (a) Find z s.t. $e^z = 1+i$.

$$\text{Note: } 1+i = \sqrt{2} \left(\frac{1+i}{\sqrt{2}} \right) = \sqrt{2} e^{i\pi/4}$$

$$\text{ie. } e^x = \sqrt{2} \quad \text{and} \quad y = \frac{\pi}{4} + 2\pi n, \quad n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{1}{2} \ln 2$$

$$\text{Hence, } z = \frac{1}{2} \ln 2 + i \left(\frac{\pi}{4} + 2\pi n \right), \quad n \in \mathbb{Z}$$

(b) $\frac{d}{dz}(e^z) = e^z$. Method 1: Direct from the definition.

$$\lim_{h \rightarrow 0} \frac{e^{z+h} - e^z}{h} = e^z \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

$$\begin{aligned} \text{Recall that } e^h &= \sum_{n=0}^{\infty} \frac{1}{n!} h^n \Rightarrow e^h - 1 = \sum_{n=1}^{\infty} \frac{1}{n!} h^n = h \sum_{n=1}^{\infty} \frac{1}{n!} h^{n-1} \\ &\Rightarrow \frac{e^h - 1}{h} = \sum_{n=1}^{\infty} \frac{1}{n!} h^{n-1} = 1 + h \sum_{n=2}^{\infty} \frac{1}{n!} h^{n-2} \end{aligned}$$

$$\text{Since } \left| h \sum_{n=2}^{\infty} \frac{1}{n!} h^{n-2} \right| \leq |h| \sum_{n=2}^{\infty} \frac{1}{n!} |h|^{n-2} \leq |h| \sum_{n=2}^{\infty} \frac{1}{(n-2)!} |h|^{n-2} = |h| e^{|h|}$$

$$\text{we conclude } \lim_{h \rightarrow 0} h \sum_{n=2}^{\infty} \frac{1}{n!} h^{n-2} = 0 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

Hence, the derivative exists for all z and is equal to e^z .

Method 2: Cauchy-Riemann Equations

$$f(z) = e^z = e^x \cos y + i e^x \sin y$$

$$u = e^x \cos y$$

$$v = e^x \sin y$$

$$u_x = e^x \cos y$$

$$v_x = e^x \sin y$$

$$u_y = -e^x \sin y$$

$$v_y = e^x \cos y$$

u, v are continuously differentiable $\forall (x, y) \in \mathbb{C}$ and satisfy the C-R eq.

The Complex Logarithm

For real numbers we can find the logarithm as the inverse of the exponential function.

$$\exp x: \mathbb{R} \rightarrow (0, \infty)$$

$$\log x: (0, \infty) \rightarrow \mathbb{R}$$

$$\text{s.t. } \exp(\log(x)) = \log(\exp(x)) = x.$$

or equivalently $y = e^x \Leftrightarrow x = \log(y)$.

However, for complex z this is complicated as e^z is not injective.

Indeed, if $e^w = z$ and $w = u + iv$

$$\Rightarrow e^u e^{iv} = z$$

$$\Rightarrow e^u = |z| \quad \text{and} \quad v = \arg(z) + 2\pi n, \quad n \in \mathbb{Z}.$$

$$\hookrightarrow u = \ln|z|$$

$$\text{So } e^w = z \Leftrightarrow w = \ln|z| + i(\arg(z) + 2\pi n) \quad n \in \mathbb{Z}.$$

There are two ways to interpret this statement.

$$(1) \quad \log z = \ln|z| + i(\arg(z) + 2\pi n) \quad n \in \mathbb{Z}$$

ie. $\log z$ is a multivalued function.

(2) We have a choice about the value of $\log z$ depending on the context in which we find it.

Note: by the definition above $e^{\log z} = z$ for any $z \neq 0$.

But $\log(e^z) \neq z$ for all z . Indeed

$$\log(e^z) = \log(e^x e^{iy}) = x + i(y + 2\pi n), \quad n \in \mathbb{Z}.$$

$$= z + 2\pi n i, \quad n \in \mathbb{Z}$$

To remove this ambiguity we will define the principal value of the logarithm as

$$\text{Log } z = \ln |z| + i \underbrace{\arg(z)}_{\in (-\pi, \pi)}.$$

Examples: (a) $\log 1 = \ln 1 + i(0 + 2\pi n), \quad n \in \mathbb{Z}$
 $= 2\pi i n, \quad n \in \mathbb{Z}$

(b) $\log(-1) = \ln|-1| + i(\pi + 2\pi n), \quad n \in \mathbb{Z}.$

(c) $\log(2-2i) = \ln|2\sqrt{2}| + i(-\frac{\pi}{4} + 2\pi n), \quad n \in \mathbb{Z}$

Branches and Derivatives

Question: Is the logarithm analytic?

Suppose we write $\log z = \ln|z| + i\theta$ where we choose $\theta \in (\alpha, \alpha + 2\pi)$.

We want to check the Cauchy-Riemann equations. However, our function is very simple in polar coordinates. Using the change of variables

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$

the Cauchy-Riemann equations are:

$$r u_r = v_\theta, \quad u_\theta = -r v_r.$$

These we can easily check as:

$$u(r, \theta) = \ln r \qquad v(r, \theta) = \theta$$

$$u_r = \frac{1}{r} \Rightarrow r u_r = 1 \qquad v_r = 0 \Rightarrow r v_r = 0$$

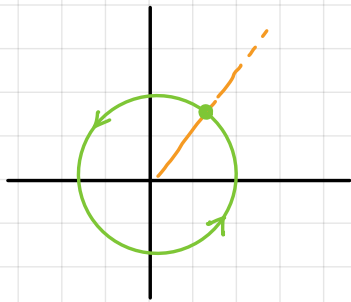
$$u_\theta = 0 \qquad v_\theta = 1.$$

By continuity of the partial derivatives, we conclude that $\log z$ is continuous and analytic for all $z \neq 0$ and $\arg(z) \in (\alpha, \alpha + 2\pi)$.

Moreover, we can use

$$\frac{d}{dz} \log z = e^{-i\theta} (u_r + i v_r) = e^{-i\theta} \left(\frac{1}{r} + i0 \right) = \frac{1}{r e^{i\theta}} = \frac{1}{z}.$$

But what happens for arguments near the boundary α :



$$\text{Let } z(t) = r_0 e^{i(\alpha+t)} \text{ for } t \in [0, 2\pi].$$

$$\text{Then } \log(z(t)) = \ln r_0 + i(\alpha+t)$$

$$\Rightarrow \log(z(0)) = \ln r_0 + i\alpha$$

$$\text{but } \log(z(2\pi)) = \ln r_0 + i(\alpha+2\pi)$$

By choosing a fixed single value for the logarithm, we have forced the function to be discontinuous along the line $\{\theta = \alpha\}$

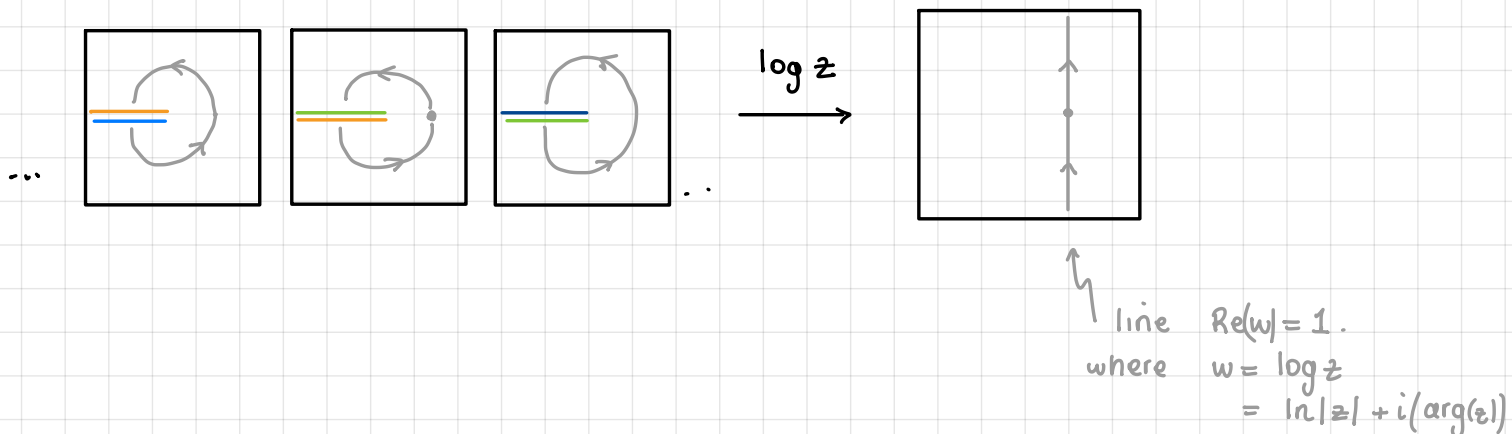
The fact that the logarithm is multi-valued is an essential property of the function.

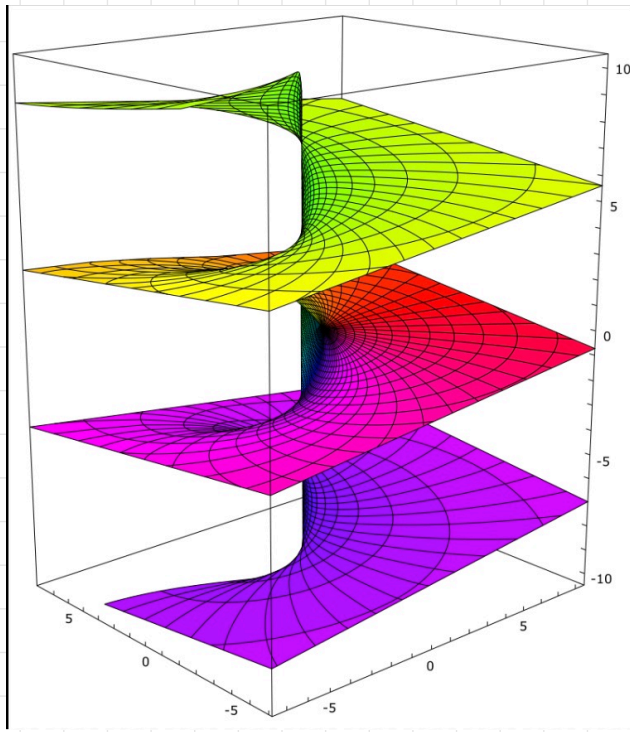
If we do not want to accept that the function is discontinuous, we can instead define it slightly differently so that the multi-values are instead found in the input space.

Since we have

$$\log z = \ln|z| + i(\arg(z) + 2\pi n) \quad n \in \mathbb{Z}$$

we imagine taking infinitely many copies of the complex plane $\mathbb{C}$ and gluing them along the branch cut $\{\theta = \pi\}$.





← Visualization of the new domain (from Wikipedia).

The new domain is called a Riemann surface, and is formed by gluing copies of the complex plane along the branch cut or line of discontinuities of the function.

Each individual copy of the complex plane, and corresponding choice of value of the function, is called a branch.

Points on the branch cuts are singular points of the function and any point that is on any choice of branch cut is called a branch point.